

Mucosal TL1A expression predicts response to anti-TL1A therapy in ulcerative colitis: a biomarker analysis of the AURORA programme

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Background

Anti-TL1A therapy is effective in a subset of patients with moderate-to-severe ulcerative colitis, but no predictive biomarker is established. We assessed whether baseline mucosal TL1A expression is associated with clinical remission.

Methods

Baseline biopsies from 402 AURORA-1 participants randomised to zoltarimab were analysed by RNA in situ hybridisation. Patients were grouped into tertiles of mucosal TL1A expression. The primary endpoint was clinical remission at week 12.

Results

Clinical remission at week 12 was achieved by 54.9% of patients in the highest expression tertile versus 21.4% in the lowest (difference 33.5%, 95% CI 22.1–44.9). The association persisted after adjustment for prior advanced therapy exposure and baseline Mayo score.

TL1A tertile	n	Clinical remission wk12	Endoscopic improvement wk12
Low	134	21.4%	29.9%
Intermediate	134	37.3%	44.0%
High	134	54.9%	61.2%

Conclusions

Baseline mucosal TL1A expression is strongly associated with response to zoltarimab. Prospective validation in a biomarker-stratified trial is warranted. A companion diagnostic is in development.

Fictional abstract created for training purposes. Not real clinical data. Disclosures: all authors are fictional.